CHAPTER 11

REPLACEMENT AND OTHER AXIOMS

We have just about reached one of the goals we set in Chapter **4**, which was to prove all the "naive" results of Chapter **2** from the axioms of Zermelo. Only a couple of minor points remain, but they are significant: they will reveal that Zermelo's axioms are not sufficient and must be supplemented by stronger principles of set construction. Here we will formulate and add to the axiomatic theory **ZDC** the AXIOM OF REPLACEMENT discovered in the early 1920's, a principle of set construction no less plausible than any of the constructive axioms (**I**) – (**VI**) but powerful in its consequences. We will also introduce and discuss some additional principles which are often included in axiomatizations of set theory. Using only a weak consequence of Replacement, we will construct the LEAST ZERMELO UNIVERSE $\mathcal{Z}$, a remarkably simple set which contains the natural numbers, Baire space, the real numbers and all the significant objects of study of classical mathematics. Everything we have proved so far can be interpreted as if $\mathcal{Z}$ comprised the entire universe of mathematical objects, yet $\mathcal{Z}$ is just a set—and a fairly small, easy to comprehend set, at that! Our main purpose in this chapter is to understand the Axiom of Replacement by investigating its simplest and most direct consequences. The real power of this remarkable proposition will become apparent in the next chapter.

According to (2) of **2.16**, if A is a countable set and for each $n \geq 2$,

$$A^n = \underbrace{A \times \cdots \times A}_{n \text{ times}},$$

then the union $\bigcup_{n=2}^{\infty} A^n$ is also countable. The obvious way to prove this from the axioms is to define first the sets A^n by the recursion

$$\begin{aligned} f(0) &= A \times A, \\ f(n+1) &= f(n) \times A, \end{aligned} \tag{11-1}$$

so that $f(n) = A^{n+2}$ and

$$\bigcup_{n=0}^{\infty} A^{n+2} = \bigcup f[\mathbb{N}]. \tag{11-2}$$

Cantor's basic **2.10** implies first (by induction) that each $f(n) = A^{n+2}$ is countable, and then that their union $\bigcup_{n=0}^{\infty} A^{n+2}$ must also be countable. Is there an error? Certainly not in the proof by induction, which is no different than many others like it. There is a problem, however, with the recursive definition (11-1) which cannot be justified by the Recursion Theorem **5.6** as it stands. To apply **5.6** we need a set E, a function $h : E \to E$ on E and some $a \in E$, which then determine a unique $f : \mathbb{N} \to E$ satisfying

$$\begin{aligned} f(0) &= a, \\ f(n+1) &= h(f(n)). \end{aligned} \tag{11-3}$$

In the case at hand there is no obvious E which contains A and all its products A^n, and instead of a function h, we have the operation

$$h(X) =_{\text{df}} X \times A, \tag{11-4}$$

which associates with each set X its product $X \times A$ with the given set A. To justify definition (11-1), we need a recursion theorem which validates recursive definitions of the form (11-3), for every object a and (unary) definite operation h. It looks quite innocuous, only a mild generalization of the Recursion Theorem—and it is just that—but in fact such a result cannot be established rigorously on the basis of the Zermelo axioms.

11.1. (**VIII**) **Replacement Axiom**. *For each set A and each unary definite operation H, the image*

$$H[A] =_{\text{df}} \{H(x) \mid x \in A\}$$

of A by H is a set.

As a construction principle for sets, the Replacement Axiom is almost obvious, as plausible on intuitive grounds as the Separation Axiom. If we already understand A as a completed totality and H associates in a definite and unambiguous manner an object with each $x \in A$, then we can "construct" the image $H[A]$ by "replacing" each $x \in A$ by the corresponding $H(x)$.

11.2. Axiomatics. The axiomatic system **ZFDC** of **Zermelo-Fraenkel set theory** with **Dependent Choices** comprises the axioms of **ZDC** and the Replacement Axiom (**VIII**), symbolically

$$\mathbf{ZFDC} = \mathbf{ZDC} + \text{Replacement} = (\mathbf{I}) - (\mathbf{VIII}).$$

From now on we will use all the axioms of **ZFDC** without explicit mention and we will continue to annotate by the mark (**AC**) the results whose proof requires the full Axiom of Choice.[29]

Mostly we have used simple definite operations up until now, those directly supplied by the axioms like $\mathcal{P}(A)$ and $\bigcup \mathscr{E}$ and explicit combinations of them, e.g., the Kuratowski pair $(x, y) =_{\text{df}} \{\{x\}, \{x, y\}\}$. Once we assume the Axiom of Replacement, however, definite operations come into center stage and we will need to deal with some which are not so simply defined. We describe in the next, trivial Proposition the basic method of definition we will use, primarily to point attention to it.

[29]The Axiom of Replacement was introduced independently by Thoralf Skolem and Abraham Fraenkel in the early 1920s.

11.3. Proposition. *Suppose C and P are definite conditions of n and $n+1$ arguments, respectively, assume that*

$$(\forall \vec{x})[C(\vec{x}) \Longrightarrow (\exists! w)P(\vec{x}, w)], \tag{11-5}$$

and let

$$F(\vec{x}) =_{\text{df}} \begin{cases} \text{the unique } w \text{ such that } P(\vec{x}, w), & \text{if } C(\vec{x}), \\ \emptyset, & \text{otherwise.} \end{cases} \tag{11-6}$$

The n-ary operation F is definite.

In practice, we will appeal to this observation by setting

$$F(\vec{x}) =_{\text{df}} \text{the unique } w \text{ such that } P(\vec{x}, w) \quad (C(\vec{x})), \tag{11-7}$$

after we verify (11-5), without specifying the irrelevant value of F outside the domain we care about. The Axiom of Replacement often comes into the proof of (11-5).

11.4. Exercise. *For each unary definite operation F, the operation*

$$G(X) =_{\text{df}} F[X] = \{F(x) \mid x \in X\} \quad (\text{Set}(X))$$

is also definite.

The next fundamental consequence of the Replacement Axiom generalizes the Transfinite Recursion Theorem in two ways: by allowing a definite operation instead of just a function in the statement, and by replacing the given well ordered set by an arbitrary grounded graph. The second generalization does not require the Replacement Axiom, Problem **x8.11**.

11.5. Grounded Recursion Theorem. *For each grounded graph G with edge relation $\to$ and each binary definite operation H, there exists exactly one function $f : G \to f[G]$ which satisfies the identity*

$$f(x) = H(f \restriction \{y \in G \mid x \to y\}, x).$$

PROOF. As in the proof of **7.24**, we first show a lemma which gives us a set of approximations of the required function. Instead of the initial segments of G (which do not make much sense for an arbitrary graph), these approximations are defined here on downward closed subsets of G. Recall the definition of the transitive closure of a graph $\Rightarrow_G$ given in **6.34**; we will skip all the subscripts in what follows, since only the single graph G is involved in the argument, and we will also use the inverse arrows,

$$u \leftarrow t \iff_{\text{df}} t \to u \iff u \text{ is immediately below } t, \tag{11-8}$$

$$x \Leftarrow t \iff_{\text{df}} t \Rightarrow x \iff x \text{ is (on some path) below } t. \tag{11-9}$$

Lemma. *For each node $t \in G$, there exists exactly one function σ with domain the set $\{x \in G \mid x \Leftarrow t\}$ which satisfies the identity*

$$\sigma(x) = H(\sigma \restriction \{y \in G \mid y \leftarrow x\}, x) \quad (x \Leftarrow t). \tag{11-10}$$

Proof. Suppose, towards a contradiction, that t is a minimal node of G where the Lemma fails. Thus, for each $u \leftarrow t$, there is exactly one function σ_u such that

$$\sigma_u(x) = H(\sigma_u \restriction \{y \in G \mid y \leftarrow x\}, x) \quad (x \Leftarrow u). \tag{11-11}$$

First we notice that

$$[x \Leftarrow u \leftarrow t \,\&\, x \Leftarrow v \leftarrow t] \Longrightarrow \sigma_u(x) = \sigma_v(x); \tag{11-12}$$

because if x were minimal in G where (11-12) failed, then

$$\begin{aligned} \sigma_u(x) &= H(\sigma_u \restriction \{y \in G \mid y \leftarrow x\}, x) && \text{by (11-11)},\\ &= H(\sigma_v \restriction \{y \in G \mid y \leftarrow x\}, x) && \text{by the choice of } x,\\ &= \sigma_v(x) && \text{by (11-11) for } \sigma_v. \end{aligned}$$

The operation $u \mapsto \sigma_u$ which assigns this σ_u to each $u \leftarrow t$ is definite, so by the Axiom of Replacement its image is a set and we can set

$$\sigma_1 =_{\text{df}} \bigcup \{\sigma_u \mid u \leftarrow t\};$$

this σ_1 is a function by (11-12), and by the definition,

$$\sigma_1(x)\downarrow \iff (\exists u)[x \Leftarrow u \leftarrow t].$$

By another application of the Replacement Axiom,

$$\sigma_2 =_{\text{df}} \{(v, H(\sigma_v \restriction \{x \mid x \leftarrow v\}), x) \mid v \leftarrow t \,\&\, \neg(\exists u)[v \Leftarrow u \leftarrow t]\}$$

is also a set, and by its definition it is a function with domain disjoint from that of σ_1. Thus

$$\sigma =_{\text{df}} \sigma_1 \cup \sigma_2$$

is a function, and

$$\begin{aligned} \sigma(x)\downarrow &\iff (\exists u)[t \to u \,\&\, u \Rightarrow x] \vee t \to u\\ &\iff t \Rightarrow x \quad (\text{by } \mathbf{6.35}). \end{aligned}$$

Moreover, σ satisfies (11-10) because σ_1 and σ_2 do. Finally, the same argument by which we proved (11-12) shows that no more than one σ with domain $\{x \in G \mid x \Leftarrow t\}$ can satisfy (11-10), and that completes the proof of the Lemma. ⊣ (Lemma)

To prove the Theorem, we apply the Lemma as in **7.24** to "the successor graph"

$$\begin{aligned} \mathrm{Succ}(G) &=_{\text{df}} G \cup \{t^*\},\\ x \to_{\mathrm{Succ}(G)} y &\iff_{\text{df}} x \to y \vee [x = t^* \,\&\, y \in G], \end{aligned}$$

which has just one more node than G, at the top. ⊣

11.6. Corollary. (1) *For each well ordered set U and each binary definite operation H, there exists exactly one function $f : U \to f[U]$ which satisfies the identity*

$$f(x) = H(f \restriction \mathbf{seg}(x), x) \quad (x \in U). \tag{11-13}$$

(2) *For each object a and each definite unary operation F, there exists a unique sequence $(n \mapsto a_n)$ which satisfies the identities*

$$a_0 = a, \quad a_{n+1} = F(a_n) \quad (n \in \mathbb{N}). \tag{11-14}$$

We call $(n \mapsto a_n)$ the **orbit** of a under F.

PROOF. For (1) we apply **11.5** to the graph $(\mathrm{Field}(U), >_U)$, and for (2) to the graph $(\mathbb{N}, \to)$, where

$$n \to m \iff_{\mathrm{df}} n = m + 1. \qquad \dashv$$

11.7. Exercise. *Which definite operation H do we use to prove* (2) *of the Corollary?*

The orbit of a set A by the unionset operation reveals the hidden structure of A under the membership relation by exposing the members of A, the members of the members of A, the members of those, etc. ad infinitum.

11.8. Definition. A class or set M is **transitive** if $\bigcup M \subseteq M$, equivalently

$$(\forall x \in M)(\forall t \in x)[t \in M],$$

or just $x \in M \Longrightarrow x \subseteq M$.

11.9. Exercise. *The sets* $\emptyset$, $\{\emptyset, \{\emptyset\}\}$, $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$, *the set*

$$\mathbb{N}_0 = \{\emptyset,\ \{\emptyset\},\ \{\{\emptyset\}\},\ \dots\} \tag{11-15}$$

postulated by the Axiom of Infinity and every class whose members are all atoms are transitive.

11.10. Transitive Closure Theorem. *Every set A is a member of some transitive set M, in fact, there is a least* (*under* $\subseteq$) *transitive set $M = \mathrm{TC}(A)$ such that $A \in \mathrm{TC}(A)$.* We call $\mathrm{TC}(A)$ the **transitive closure** of A.

PROOF. By (2) of **11.6**, there is a unique sequence $n \mapsto \mathrm{TC}_n(A)$ which satisfies the identities

$$\begin{aligned} \mathrm{TC}_0(A) &= \{A\}, \\ \mathrm{TC}_{n+1}(A) &= \bigcup \mathrm{TC}_n(A), \end{aligned} \tag{11-16}$$

and we set

$$\mathrm{TC}(A) =_{\mathrm{df}} \bigcup_n \mathrm{TC}_n(A). \tag{11-17}$$

Clearly $A \in \mathrm{TC}(A)$ and $\mathrm{TC}(A)$ is transitive, because

$$u \in \mathrm{TC}_n(A) \Longrightarrow u \subseteq \bigcup \mathrm{TC}_n(A) = \mathrm{TC}_{n+1}(A).$$

If M is transitive and $A \in M$, then $\mathrm{TC}_0(A) = \{A\} \subseteq M$, and by induction

$$\mathrm{TC}_n(A) \subseteq M \Longrightarrow \mathrm{TC}_{n+1}(A) = \bigcup \mathrm{TC}_n(A) \subseteq \bigcup M \subseteq M,$$

so that in the end $\mathrm{TC}(A) = \bigcup_n \mathrm{TC}_n(A) \subseteq M$. $\dashv$

11.11. Exercise. *If A is transitive, then* $\mathrm{TC}(A) = A \cup \{A\}$.

To understand better the remark about "revealing the hidden $\in$-structure" of A, consider the following natural concepts.

11.12. Definition. A set A is *hereditarily free of atoms* or **pure** if it belongs to some transitive set which contains no atoms; equivalently, if $\mathrm{TC}(A)$ contains no atoms. A set A is **hereditarily finite** if it belongs to some transitive, finite set; equivalently, if $\mathrm{TC}(A)$ is finite. A set A is **hereditarily countable** if it belongs to some transitive, countable set; equivalently, if $\mathrm{TC}(A)$ is countable.

The point of the definitions is that $\{\{a\}\}$ is a set but not a pure set if a is an atom, because we need a to construct it; $\{\mathbb{N}\}$ is finite but not hereditarily finite because we need all the natural numbers to construct it; $\{\mathcal{N}\}$ is countable but not hereditarily countable because we need to "collect into a whole" an uncountable collection of objects in $\mathcal{N}$ before we can construct the singleton $\{\mathcal{N}\}$ by one final, trivial act of collection. Put another way, $\{\mathcal{N}\}$ is not hereditarily countable because "its concept involves" an uncountable infinity of objects, the members of its sole member $\mathcal{N}$.

11.13. Exercise. *The Principle of Purity* **3.25** *is equivalent to the assertion that every set is pure.*

11.14. Exercise. *A transitive set is hereditarily finite if it is finite, and hereditarily countable if it is countable.*

Next we consider the closure of a set under both the unionset and powerset operations.

11.15. Theorem (Basic Closure Lemma). *For each set I and each natural number n, let $M_n = M_n(I)$ be the set defined by the recursion*

$$M_0 = I, \quad M_{n+1} = M_n \cup \bigcup M_n \cup \mathcal{P}(M_n). \tag{11-18}$$

The **basic closure** *of I is the union*

$$M = M(I) =_{\mathrm{df}} \bigcup\nolimits_{n=0}^{\infty} M_n(I), \tag{11-19}$$

and it has the following properties.

(1) *M is a transitive set which contains $\emptyset$ and I, it is closed under the pairing $\{x, y\}$, unionset $\bigcup\mathscr{E}$ and powerset $\mathcal{P}(A)$ operations and it contains every subset of each of its elements.*

(2) *M is the least (under $\subseteq$) transitive set which contains I and is closed under $\{x, y\}$, $\bigcup\mathscr{E}$ and $\mathcal{P}(X)$.*

(3) *If I is pure and transitive, then each M_n is a pure, transitive set and satisfies*

$$M_{n+1} = \mathcal{P}(M_n). \tag{11-20}$$

As a consequence, M is a pure, transitive set.

PROOF. (1) By the definition, $\emptyset, I \in M_1 \subseteq M$. If $x, y \in M$, then from the obvious $M_n \subseteq M_{n+1}$, there exists some m such that $\{x, y\} \subseteq M_m$, so $\{x, y\} \in M_{m+1}$. The key inclusion for the remaining claims is

$$x \in M_n \Longrightarrow x \subseteq \bigcup M_n \subseteq M_{n+1} \subseteq M,$$

which implies immediately that M is transitive. It also implies

$$x \in M_n \Longrightarrow \bigcup x \subseteq \bigcup M_{n+1} \subseteq M_{n+2},$$

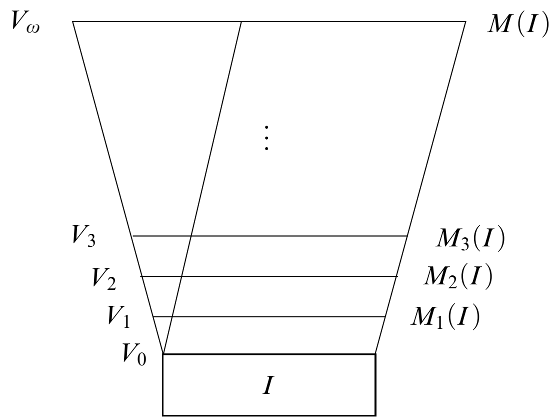

FIGURE **11.1**. Logarithmic[30] renditions of $M(I)$ and V_ω.

so $x \in M_n \Longrightarrow \bigcup x \in M_{n+3} \subseteq M$ and M is closed under $\bigcup x$. The same argument shows that M is closed under $\mathcal{P}(X)$ and the last assertion follows by this closure and transitivity.

(2) If M' is closed under $\{x, y\}$ and $\bigcup \mathscr{E}$, then it is also closed under $A \cup B = \bigcup\{A, B\}$; and if M' is also closed under $\mathcal{P}(X)$, then a simple induction shows that $M_n \in M'$ for each n, and so $M \subseteq M'$ by the transitivity of M'.

(3) If I is transitive with no atoms, then every M_n is transitive and has no atoms by a trivial induction on n. This implies that M is a transitive set with no atoms and hence pure, but also that $M_n \cup \bigcup M_n \subseteq \mathcal{P}(M_n)$, so that $M_{n+1} = \mathcal{P}(M_n)$. ⊣

11.16. Exercise. *True or false*: *for every transitive set X, $X \subseteq \mathcal{P}(X)$.*

11.17. Exercise. *If $I \subseteq J$, then for each n, $M_n(I) \subseteq M_n(J)$, and hence*

$$I \subseteq J \Longrightarrow M(I) \subseteq M(J).$$

11.18. The grounded, pure, hereditarily finite sets. The least basic closure is that of the empty set, $M(\emptyset) \subseteq M(I)$, for every I. In the classical notation (which we will explain in the next chapter), $M_n(\emptyset) = V_n$, so that each V_n and their union are determined by the identities

$$V_0 = \emptyset, \quad V_{n+1} = \mathcal{P}(V_n), \quad V_\omega =_{\mathrm{df}} \bigcup_{n=0}^{\infty} V_n = M(\emptyset). \tag{11-21}$$

For example, $\emptyset \in V_1$, $\{\emptyset\} \in V_2$ and $\{\{\emptyset\}, \{\{\emptyset\}\}\} \in V_4$! These sets are pictured on the left in Figure **11.1**. The set V_ω is grounded, pure and transitive, each V_n is finite by an easy induction, so every set in V_ω is grounded, pure and hereditarily finite, and V_ω itself is countable. These are the sets which can

[30]Picturing universes of sets by cones like this is traditional but misleading: the successive powersets grow hyperexponentially in size, so it would be more accurate to draw a cone with curved, hyperexponential sides.

be constructed "from nothing" (literally, the empty set) by iterating any finite number of times the operation of collecting into a whole (putting between braces) some of the objects already constructed.

The closure properties of $M(I)$ itemized in (1) of **11.15** are precisely those required of the universe $\mathcal{W}$ by axioms (**II**) – (**V**), as we discussed them in **3.26**, and if the set $\mathbb{N}_0$ of (11-15) demanded by the Axiom of Infinity (**VI**) is a subset of I, we also have $\mathbb{N}_0 \in M(I)$. Notice also that since $M(I)$ is transitive, for $A, B \in M(I)$,

$$A \neq B \Longrightarrow (\exists t \in M(I))[t \in (A \setminus B) \cup (B \setminus A)], \tag{11-22}$$

which says of $M(I)$ what the Axiom of Extensionality demands of $\mathcal{W}$ by (3-12). This suggests that if we take "object" to mean "member of $M(I)$", for any $I \supseteq \mathbb{N}_0$, then we can reinterpret every proof from the axioms (**I**) – (**VI**) as an argument about the members of $M(I)$ instead of all objects, in the end proving a theorem about $M(I)$ instead of $\mathcal{W}$. It is an important idea, worth abstraction and a name.

11.19. Definition. A transitive class M is a **Zermelo universe** if it is closed under the pairing $\{x, y\}$, unionset $\bigcup \mathscr{E}$ and powerset $\mathcal{P}(X)$ operations, and contains the set $\mathbb{N}_0$ defined by (11-15). The least Zermelo universe is $\mathcal{Z} = M(\mathbb{N}_0)$, determined by the identities

$$\mathcal{Z}_0 = \mathbb{N}_0, \quad \mathcal{Z}_{n+1} = \mathcal{P}(\mathcal{Z}_n), \quad \mathcal{Z} = \bigcup_{n=0}^{\infty} \mathcal{Z}_n. \tag{11-23}$$

11.20. Exercise. *The class $\mathcal{W}$ of all objects is a Zermelo universe. Every Zermelo universe contains the empty set as well as every subset of each of its members.*

A Zermelo universe M is a *model* of the axioms (**I**) – (**VI**), and a very special model at that, since it interprets standardly the basic relations of membership and sethood—it only restricts the domain of objects in which we interpret propositions. The claim that logical consequences of (**I**) – (**VI**) are true in every Zermelo universe is called a *metatheorem*, a theorem about theorems. To make general results of this type completely precise and prove them rigorously requires concepts from *Mathematical Logic*. In specific instances, however, lemma by lemma and proposition by proposition, it is quite simple to see what the specific consequence of the axioms means for an arbitrary domain of objects and to verify it in every Zermelo universe: this is because, in fact, we have been using the axioms as closure properties of the universe, about which we have assumed nothing more but that it satisfies them.

11.21. Proposition. *Every Zermelo universe M is closed under the Kuratowski pair operation (x, y) defined in* (4-1), *as well as the Cartesian product $A \times B$, function space $(A \to B)$, and partial function space $(A \rightharpoonup B)$ operations, provided these are defined using the Kuratowski pair. In addition, if $A \in M$ and $\sim$ is an equivalence relation on A, then $\sim$ and the quotient $[\![A/{\sim}]\!]$ defined in* **4.12** *are also in M.*

PROOF. The Kuratowski pair $(x, y) = \{\{x\}, \{x, y\}\}$ of any two members of M is obtained by taking unordered pairs twice, so it is certainly in M. If $A, B \in M$, then $A \cup B = \bigcup\{A, B\}$, and by the proof of **4.2**,

$$A \times B \subseteq \mathcal{P}(\mathcal{P}(A \cup B)) \in M,$$

so $A \times B \in M$. The remaining claims are proved similarly. ⊣

11.22. Exercise. *Every Zermelo universe M contains a Peano system as defined in* **5.1**.

11.23. Proposition. (1) *The Axiom of Dependent Choices is true in every Zermelo universe M, in the following sense: if $a \in A \in M$, $P \subseteq A \times A$, $P \in M$ and $\mathbb{N} \in M$ is a system of natural numbers in M, then*

$$\begin{aligned} &a \in A \,\&\, (\forall x \in A)(\exists y \in A)P(x, y) \\ &\quad \Longrightarrow (\exists f : \mathbb{N} \to A)[f \in M \,\&\, f(0) = a \,\&\, (\forall n \in \mathbb{N})P(f(n), f(n+1))]. \end{aligned}$$

(2) (**AC**) *The Axiom of Choice is true in every Zermelo universe M, in the following sense, following* **8.4**: *for every family $\mathscr{E} \in M$ of non-empty and pairwise disjoint sets, there exists some set $S \in M$ which is a choice set for $\mathscr{E}$, i.e.,*

$$S \subseteq \bigcup \mathscr{E}, \quad (\forall X \in \mathscr{E})(\exists u)[S \cap X = \{u\}].$$

PROOF. (1) The hypothesis of the implication to be proved implies by **DC** that there exists some function $f : \mathbb{N} \to A$ such that $f(0) = a$ and for every $n \in \mathbb{N}$, $P(f(n), f(n+1))$. Since $(\mathbb{N} \to A) \in M$, we also have $f \in M$ by transitivity.

Part (2) is proved similarly. ⊣

Although we chose specific versions of the choice principles to simplify these arguments, their numerous equivalents are also true in every Zermelo universe M. This can be verified directly, or by observing that the equivalence proofs we have given can be "carried out within M".

11.24. Exercise. (**AC**) *If M is any Zermelo universe, $A, B \in M$, and P is any binary definite condition, then*

$$\begin{aligned} &(\forall x \in A)(\exists y \in B)P(x, y) \\ &\qquad \Longrightarrow (\exists f \in M)[f : A \to B \,\&\, (\forall x \in A)P(x, f(x))]. \end{aligned} \tag{11-24}$$

11.25. The least Zermelo universe $\mathcal{Z}$. Let us concentrate on the least Zermelo universe $\mathcal{Z}$, to focus the argument. It is a pure, transitive set, constructed by starting with the simple set $\mathbb{N}_0$ and iterating the powerset operation infinitely many times, much as we construct the natural numbers starting with 0 and iterating infinitely many times the successor operation. We can think of the sets in $\mathcal{Z}$ as precisely those objects whose existence is guaranteed by the axioms (**I**) – (**VI**). The natural interpretations of **DC** and **AC** are also true in $\mathcal{Z}$, the latter under the assumption that **AC** is true in $\mathcal{W}$. Using the closure properties of Zermelo universes already established and looking back at Chapters **5**, **6** and **10** and ahead at Appendix **A**, we can verify that $\mathcal{Z}$ contains not only the specific system of natural numbers $\mathbb{N}$ we constructed in Chapter **5**, but also the Baire space $\mathcal{N}$ defined from this $\mathbb{N}$ and the specific systems of rational and real numbers constructed in Appendix **A**. By the uniqueness results, any one of these systems is as good as any other, so we can say that $\mathcal{Z}$ contains the integers, Baire space, the rationals and the reals.

Combining these remarks with some knowledge of classical mathematics, it is not hard to give a convincing argument that *all the objects studied in classical algebra, analysis, functional analysis, topology, probability, differential equations, etc. can be found (to within isomorphism) in* $\mathcal{Z}$. Many fundamental objects of abstract set theory are also in $\mathcal{Z}$, all we have constructed before this chapter in developing the theory of inductive posets, well ordered sets, etc. In slogan form: **we can develop classical mathematics and all the set theory needed for it as if all mathematical objects were members of $\mathcal{Z}$.**

The same can be said of every Zermelo universe, of course, but the concrete, simple definition of $\mathcal{Z}$ makes it possible to analyze its structure and investigate the special properties of its members. For example, no set which is a member of itself belongs to $\mathcal{Z}$: because no $X \in \mathbb{N}_0$ satisfies $X \in X$ (easily), and if n were least such that some $X \in X \in \mathcal{Z}_{n+1}$, then $X \in X \subseteq \mathcal{Z}_n$ by the definition, so $X \in \mathcal{Z}_n$, contradicting the choice of n. This looks good, we had some trouble with sets which belong to themselves. Actually, the iterative construction of $\mathcal{Z}$ ensures a much stronger regularity property for its members, discovered by von Neumann.

11.26. Definition. An object x is **ill founded** if it is the beginning of a descending $\in$-chain, i.e., if there exists a function $f : \mathbb{N} \to E$ such that

$$x = f(0) \ni f(1) \ni f(2) \ni \cdots .$$

Objects which are not ill founded are **well founded** or **grounded**. If $X \in X$, then $X \ni X \ni X \ni \cdots$, so X is ill founded. Problem **x11.14** gives a simple characterization of ill founded sets directly in terms of the $\in$ relation, which suggests that ill foundedness is a generalization of self-membership.

11.27. Exercise. *Atoms are grounded, as is $\emptyset$ and $\mathbb{N}_0$. A set is grounded if and only if all its members are grounded, if and only if its powerset is grounded. The class of all grounded sets is transitive.*

11.28. Proposition. *If I is grounded, then so is its basic closure $M(I)$. In particular, the least Zermelo universe $\mathcal{Z}$ and all its members are grounded.*

PROOF. Assume that I is grounded, let (towards a contradiction) n be least such that M_n is ill founded and suppose that $M_n \ni x_1 \ni \cdots$ is a descending $\in$-chain. By hypothesis $n > 0$. Since $x_1 \in M_{n-1}$ and $x_1 \in y \in M_{n-1}$ contradict the choice of n, we must have $x_1 \subseteq M_{n-1}$, so $x_2 \in M_{n-1}$ and the descending $\in$-chain $M_{n-1} \ni x_2 \ni \cdots$ contradicts again the choice of n. It follows that M is also grounded, since any descending chain $M \ni x_1 \ni \cdots$ would also witness that M_n is ill founded for whatever M_n contains x_1. The consequence about $\mathcal{Z}$ follows because $\mathbb{N}_0$ is grounded. ⊣

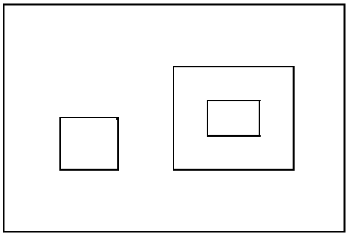

FIGURE **11.2**. $\{\emptyset, \{\emptyset\}\}$ as a disappointing gift.

In the old gag, the excited birthday boy opens up the huge box with his present, only to find inside it another box, and inside that another, and so on, until the last, tiny box is empty: his present is just the boxes. We can think of a pure, grounded set as a disappointing gift of this sort, except that each box may contain several boxes, not just one; no matter which one the birthday boy chooses to open up each time, eventually he finds the empty box, $\emptyset$. Most axiomatizations of set theory ban ill founded sets from the start by adopting the following principle proposed by von Neumann.

11.29. Principle of Foundation. *Every set is grounded.* This is also called the principle (or axiom) of *Regularity* in the literature.

It is worth putting down here an equivalent version of this Principle, which is somewhat opaque but useful.

11.30. Proposition. *The Principle of Foundation is true if and only if for every non-empty set X, there is some $m \in X$ such that*

$$m \cap X = \emptyset. \tag{11-25}$$

PROOF. Assume first the Principle of Foundation and suppose, towards a contradiction, that $X \neq \emptyset$ but no $m \in X$ satisfies (11-25). This means that for some a,

$$a \in X \;\&\; (\forall m \in X)(\exists t \in X)[t \in m],$$

and then **DC** gives us an infinite descending $\in$-chain beginning with $X \ni a$ which contradicts the hypothesis. Conversely, if the Principle of Foundation fails and some infinite descending $\in$-chain starts with some set

$$X = f(0) \ni f(1) \ni f(2) \ni \cdots,$$

then the set $f[\mathbb{N}] = \{f(0), f(1), \dots\}$ is not empty and intersects each of its members, so none of them satisfies (11-25). $\dashv$

11.31. Zermelo-Fraenkel set theory (with choice), ZFC. By far the most widely used—the "official"—system of axioms for sets is the **Zermelo-Fraenkel Theory** (with choice), which accepts the Principles of Purity **3.25** and Foundation in addition to those of **ZFDC** and the Axiom of Choice, symbolically,

$$\mathbf{ZFC} = \mathbf{ZFDC} + \mathbf{AC} + \text{Purity} + \text{Foundation}.$$

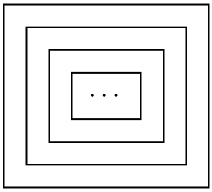

FIGURE **11.3**. Ω as the ultimately frustrating gift.

There are many and convincing arguments in favor of this industry standard, some of which we discuss immediately below. We will come back to the question in Chapter **12** and Appendix **B**, where we will also explain a few good foundational reasons for sticking with the weaker **ZFDC** in these Notes. As a practical matter, the principles of Purity and Foundation do not come up in the part of the subject we are covering, and the full **AC** is only needed rarely, so we can easily keep track of it.

11.32. Are all sets grounded? The most blatant exception to the Principle of Foundation would be a set which is its own singleton,

$$\Omega = \{\Omega\}. \tag{11-26}$$

We can think of Ω as the ultimately frustrating gift: each box has exactly one box inside it, identical with the one you just opened, and you can keep opening them forever without ever finding anything. How about sets Ω^1 and Ω^2 such that

$$\Omega^1 = \{\emptyset, \Omega^2\}, \quad \Omega^2 = \{\Omega^1\}? \tag{11-27}$$

These equations look unlikely, even bizarre, but it is not clear that our axioms rule them out. As a matter of fact they do not: we will construct in Appendix **B** some quite reasonable models of **ZFDC**+**AC** which contain some $\Omega = \{\Omega\}$ and many other sets with similar properties.

Recall the discussion about the *large* and the *small* heuristic views of the universe of objects $\mathcal{W}$ in **3.26**. The large view conceives $\mathcal{W}$ as the largest possible collection of objects which satisfies the axioms, while the small view takes it to comprise just the objects guaranteed by them.

On the large view, we have no more evidence in favor or against the Principle of Foundation now than we did back in Chapter **3**, except that we have proved all these things about sets without ever using it. But then again, we never saw a need for ill founded sets either.

On the small view, we have amassed some considerable evidence, at least for **ZDC**+**AC**, and it is all in favor of the Principle of Foundation: we now have a precise idea of what sets are "guaranteed" by the axioms of **ZDC**+**AC**, they are the members of $\mathcal{Z}$ and they are all pure and grounded. It may be argued that we did not build $\mathcal{Z}$ out of whole cloth, we worked within a "given" universe $\mathcal{W}$ of objects: in fact, we needed to assume that $\mathcal{W}$ satisfies the Axiom of Replacement in addition to the axioms of **ZDC**+**AC**. This is certainly true, but so is the obvious response to it: aside from any rigorous axiomatization, the definition of $\mathcal{Z}$ and the proofs of its basic properties can be understood intuitively, naively, and they carry considerable force of persuasion. An informal description of $\mathcal{Z}$ would have made perfect sense in Chapter **3**, as an intuitive conception of "restricted set" which justifies the axioms of **ZDC**+**AC** and the principles of Purity and Foundation. We have not been able to produce any such plausible, intuitive model of **ZDC** which contains ill founded sets from any hypotheses which do not beg the question.[31]

Could we construct simple models like $\mathcal{Z}$ for the theories **ZFDC** and **ZFC**? Let us first give them a name.

11.33. Definition. A **ZFDC-universe** is any Zermelo universe M which further satisfies the Axiom of Replacement in the following sense: for each $A \in M$ and each unary definite operation H,

$$(\forall x \in M)[H(x) \in M] \Longrightarrow H[A] = \{H(x) \mid x \in A\} \in M.$$

A **ZFC-universe** is any **ZFDC**-universe M which contains no atoms and such that every set $A \in M$ admits a well ordering in M.

The axioms of **ZFDC** assert precisely that the class $\mathcal{W}$ of all objects is a **ZFDC**-universe, and, correspondingly, the axioms of **ZFC** claim that $\mathcal{W}$ is a **ZFC**-universe.

11.34. Theorem. *The* **von Neumann class**

$$\mathcal{V} =_{\mathrm{df}} \{X \mid X \text{ is a pure, grounded set}\} \tag{11-28}$$

is a **ZFDC**-*universe*; *and if* **AC** *holds, then* $\mathcal{V}$ *is a* **ZFC**-*universe.*

PROOF. The fact that $\mathcal{V}$ is a Zermelo universe is quite trivial, most of it following from Exercise **11.27**. To verify that $\mathcal{V}$ also satisfies the Axiom of Replacement, notice that (whether $A \in \mathcal{V}$ or not), if H is unary, definite and such that for every $x \in A$, the value $H(x)$ is a pure, grounded set, then the image $H[A]$ has only pure and grounded members, so it is (easily) pure and grounded.

The second claim follows immediately, because if **AC** holds, then every $A \in \mathcal{V}$ admits a well ordering $\leq_A \in \mathcal{P}(A \times A)$ and $\leq_A \in \mathcal{V}$. ⊣

There is another, elegant and useful characterization of the pure grounded sets which follows easily from the Grounded Recursion Theorem **11.5**.

11.35. Definition. A **Mostowski surjection** or **decoration** of a graph G with edge relation $\rightarrow$ is a surjection $d : G \twoheadrightarrow d[G]$ which assigns a set to each node of G such that

$$d(x) = \{d(y) \mid y \leftarrow x\} \quad (x \in G). \tag{11-29}$$

[31]Models like $M(\mathbb{N}_0 \cup \Omega)$ beg the question, because they need some Ω with the requisite self-membership property to get started.

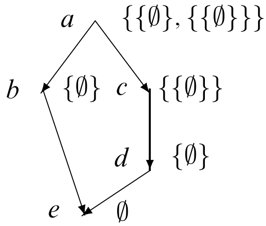

FIGURE **11.4**. Mostowski collapsing; $d_G[G] = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\{\emptyset\}, \{\{\emptyset\}\}\}\}$.

11.36. Theorem (Mostowski Collapsing Lemma). (1) *Every grounded graph G admits a unique decoration d_G, and its image $d_G[G]$ is a transitive, pure, grounded set.*

(2) *A set A is pure and grounded if and only if there exists a grounded graph G and a node $x \in G$, such that $A = d_G(x)$ for the unique decoration d_G of G.*

PROOF. (1) The existence of a unique decoration of a grounded G follows immediately from the Grounded Recursion Theorem **11.5** applied to G, with the definite operation

$$H(f) = \mathrm{Image}(f) = \{f(x) \mid f(x)\downarrow\}.$$

The image $d_G[G]$ is transitive, since if $s \in t \in d_G[G]$, then $s \in t = d_G(y)$ for some $y \in G$, and then $s = d_G(x)$ for some $x \leftarrow y$, so $s \in d_G[G]$. Since each $d_G(x)$ is a set, by (11-29), $d_G[G]$ is a transitive set with no atoms and hence pure. Finally, $d_G[G]$ is grounded, because if $x_0 \ni x_1 \ni \cdots$ were an infinite, descending $\in$-chain in it and $s_0, s_1, \ldots$ were chosen so that $d_G(s_i) = x_i$, then $s_0 \to s_1 \to \cdots$ would be an infinite descending chain in the grounded graph G.

(2) If $A = d_G(x)$ with x a node in some grounded graph, then A is a member of a transitive, pure, grounded set by (1) and hence pure and grounded. For the converse, let $G = \mathrm{TC}(A)$ be the transitive closure of A and define on it

$$x \to y \iff_{\mathrm{df}} y \in x.$$

The graph G is grounded, because $G = \mathrm{TC}(A)$ is a grounded set, Problem **x11.16**. In addition,

$$d_G(x) = x \quad (x \in G); \tag{11-30}$$

because if x were a G-minimal counterexample to (11-30), then

$$\begin{aligned} d_G(x) &= \{d_G(y) \mid y \ {}_G\!\leftarrow x\}, \\ &= \{y \mid y \leftarrow x\} && \text{by the choice of } x, \\ &= \{y \mid y \in x\} && \text{by the def. of } \to, \\ &= x && \text{because } x \text{ is a set.} \end{aligned}$$

In particular, $A = d_G(A)$, which proves (2). ⊣

The class $\mathcal{V}$ is not a set (Problem **x11.20**) and it is quite hard to find **ZFDC**-universes which are sets. See Problems **x12.43**, **x12.44** and **xB.12**.

11.37. Consistency and independence results. All the consistency and independence results we have discussed in **8.24**, **8.25**, **10.33**, **10.34**, **10.36** and **10.37** can be strengthened by adding the Axiom of Replacement to the relevant theories. This is as good a place as any to collect the most general versions of these fundamental results, which are outside the scope of these Notes.

(1) (Gödel, 1939) *The universe L of constructible sets is a model of* **ZFC**, *which further satisfies the Generalized Continuum Hypothesis* **GCH**. It follows that the Axiom of Choice **AC** cannot be refuted from the other axioms of **ZFC**, and that **GCH** cannot be refuted in **ZFC**.

(2) (Cohen, 1963) *None of the choice principles* $\mathbf{AC}_{\mathbb{N}}$, **DC** *and* **AC** *can be proved from a weaker one using the constructive axioms of Zermelo* (**I**) – (**VI**) *and the Axiom of Replacement* (**VIII**).

(3) (Cohen, 1963) *There is a model of* **ZFC** *in which the Continuum Hypothesis* **CH** *is false,* so **CH** *cannot be proved in* **ZFC**.

(4) (Solovay, 1970) *There is a model of* **ZFC** *in which every "definable", uncountable pointset has a perfect subset, and hence has cardinality* $\mathfrak{c}$. This means in particular, that we cannot define a specific pointset A and then prove in **ZFC** that it has cardinality intermediate between $\aleph_0$ and $\mathfrak{c}$.

(5) (Solovay, 1970) *There is a model of* **ZFDC** *in which every uncountable pointset has a perfect subset*, so *we cannot prove in* **ZFDC** *the existence of uncountable pointsets without perfect subsets*. Solovay's model also satisfies the Principles of Purity and Foundation.

Problems for Chapter 11

x11.1. Prove the Separation Axiom (**III**) from the remaining axioms in the group (**I**) – (**V**) and the Axiom of Replacement (**VIII**).

x11.2. For each set A, each unary, definite operation F and each binary, definite operation G, there exists a least under $\subseteq$ set $\overline{A}$ which contains A as a subset and is closed under F and G, i.e.,

$$A \subseteq \overline{A}, \quad x \in \overline{A} \Longrightarrow F(x) \in \overline{A}, \quad x, y \in \overline{A} \Longrightarrow G(x, y) \in \overline{A}.$$

(The same is true for any number of operations, of any number of arguments.)

x11.3. The Axiom of Replacement is constructively equivalent with the following proposition: for every set A and every unary definite operation F, there exists a set B which contains A and is closed under F, i.e.,

$$A \subseteq B \,\&\, F[B] \subseteq B.$$

x11.4. The Axiom of Replacement is constructively equivalent with the following proposition: for every set A and every unary definite operation F, the restriction

$$F \upharpoonright A =_{\mathrm{df}} \{(x, F(x)) \mid x \in A\}$$

of F to A is a function, i.e., a set of pairs.

x11.5. If (x, y) is a definite, binary operation which satisfies the first property of ordered pairs (OP1) in **4.1**, then it also satisfies the second, (OP2). (This cannot be proved in **ZDC+AC**, see Problem **xB.4**.)

x11.6. If $|A|$ is a definite operation which satisfies the first condition on weak cardinal assignments (C1), then it automatically also satisfies the third one, (C3). (This cannot be proved in **ZDC+AC**, see Problem **xB.8**.)

x11.7. The definite condition of functionhood defined in (4-14) satisfies the equivalence

$$\begin{aligned}\mathrm{Function}(f) \iff\ & \mathrm{Set}(f) \,\&\, (\forall w \in f)(\exists x, y)[w = (x, y)] \\ & \&\, (\forall x, y, y')[[(x, y) \in f \,\&\, (x, y') \in f] \Longrightarrow y = y'],\end{aligned}$$

i.e., f is a function exactly when it is a single-valued set of pairs. (See also Problem **xB.9**.)

x11.8. There exists a sequence $(n \mapsto \aleph_n)$ which satisfies the identities

$$\aleph_0 = |\mathbb{N}|, \quad \aleph_{n+1} = \aleph_n^+.$$

We introduced these names for the first few infinite cardinals in (9-6), but this is not the same as proving the existence of *the sequence* $(n \mapsto \aleph_n)$. (See also Problem **xB.10**.)

x11.9. Extended recursion with parameters. For every unary definite operation G and every ternary definite operation H, there exists a unary definite operation F which satisfies the identities

$$\begin{aligned}F(0, y) &= G(y), \\ F(n+1, y) &= H(F(n, y), n, y).\end{aligned}$$

x11.10. If $\mathscr{E}$ is a non-empty family of transitive sets, then the union $\bigcup \mathscr{E}$ and the intersection $\bigcap \mathscr{E}$ are also transitive.

x11.11. For every class A there exists a least, transitive class $\overline{A}$ which contains A, that is, such that $A \subseteq \overline{A}$ and for every transitive class B, $A \subseteq B \Longrightarrow \overline{A} \subseteq B$.

x11.12. The class of all pure sets is transitive, as are the classes of hereditarily finite and hereditarily countable sets.

x11.13. If $x_1 \in x_2 \in \cdots \in x_n = x_1$, then x_1 is ill founded.

x11.14. An object x is ill founded if and only if there exists some set A such that

$$x \in A \,\&\, (\forall s \in A)(\exists t \in A)[s \ni t].$$

x11.15. If Ω^1 and Ω^2 exist which satisfy (11-27), then they are distinct, hereditarily finite, pure sets.

x11.16. A set A is grounded if and only if its transitive closure $\mathrm{TC}(A)$ is grounded.

***x11.17.** A set is in V_ω if and only if it is pure, grounded and hereditarily finite. HINT. Show first that every finite, transitive, pure grounded set is in V_ω.

x11.18. For each transitive set I, let

$$J = \{x \in I \mid x \text{ is pure and grounded}\}$$

and prove that

$$M(J) = \{x \in M(I) \mid x \text{ is pure and grounded}\}.$$

x11.19. If a set $\Omega = \{\Omega\}$ exists as in (11-26), then

$$\{x \in M(\Omega) \mid x \text{ is grounded}\} = V_\omega.$$

x11.20. Prove that the class $\mathcal{V}$ of all pure, grounded sets is not a set.

11.38. Definition. A class K of atoms **supports** a set A if

$$x \in \mathrm{TC}(A)\ \&\ \mathrm{Atom}(x) \Longrightarrow x \in K,$$

and we let

$$\mathcal{W}[K] = \{x \mid x \text{ is supported by } K\}, \tag{11-31}$$
$$\mathcal{V}[K] = \{x \mid x \text{ is grounded and supported by } K\}. \tag{11-32}$$

x11.21. The class $\mathcal{W}[K]$ of sets supported by a class of atoms K is a **ZFDC**-universe.

x11.22. The class $\mathcal{V}[K]$ of grounded sets supported by a class K of atoms is a **ZFDC**-universe.

x11.23. Let $G = (\mathbb{N}, \rightarrow)$ where $m \rightarrow n \iff n < m$. Show that G is grounded and compute the image $d_G(m)$ of each m under the unique decoration of G.

x11.24. Give an example of an infinite, grounded graph G with $d_G[G] = \{\emptyset\}$.

***x11.25.** Let $G = (\mathbb{N} \setminus \{0\}, \rightarrow)$, where

$$m \rightarrow n \iff m \neq n\ \&\ n \text{ divides } m.$$

Show that G is grounded and compute the image $d_G(m)$ of each m under the unique decoration of G.

x11.26. An *extended decoration* of a graph G is any surjection $d : G \twoheadrightarrow d[G]$ such that for all $x \in G$,

$$d(x) = \begin{cases} x, & \text{if } x \text{ is an atom,} \\ \{d(y) \mid y \leftarrow x\}, & \text{if } x \text{ is a set.} \end{cases}$$

Prove that every grounded graph admits a unique extended decoration and that a (not necessarily pure) set A is grounded if and only if there exists a grounded graph G and some $x \in G$, such that $A = d(x)$.

*__x11.27. Grounded $\in$-recursion__. For each binary definite operation H, there exists a definite operation $F(t)$, such that for every grounded set x,

$$F(x) = H(F \upharpoonright x, x),$$

where the function $F \upharpoonright x = \{(t, F(t)) \mid t \in x\}$ is the restriction of the operation F to the set x.

This is a special case of the next, slightly more complex generalization of the Grounded Recursion Theorem **11.5**.

*__x11.28.__ Suppose $t < x$ is a binary definite condition which satisfies (1) for every x, the class $\{t \mid t < x\}$ is a set, and (2) there does not exist a sequence $(n \mapsto x_n)$ such that for all n, $x_{n+1} < x_n$. Prove that for every definite, binary operation H there exists another F, such that for every x,

$$F(x) = H(F \upharpoonright \{t \mid t < x\}, x),$$

where the function $F \upharpoonright \{t \mid t < x\} = \{(t, F(t)) \mid t < x\}$ is the restriction of F to the set $\{t \mid t < x\}$.